

Nima Jarahiyan

Address: Scotland, UK **Email:** n.jarahiyan@gmail.com **Mobile:** +44 (0)7803 086785 **LinkedIn:** Nima Jarahiyan

Summary

FPGA and embedded systems engineer with 7+ years of experience building high-throughput, low-latency systems for aerospace, wireless communications, and automotive applications. Work spans FPGA architecture, embedded software, high-speed interfaces, and hardware acceleration, from early design through production. Experienced team lead who brings hardware, software, and multidisciplinary teams together to solve demanding engineering problems.

Technical Expertise

FPGA, SoC, and Processors:

PolarFire SoC (5x RISC-V + Flash-Based FPGA), Kintex Ultrascale+ (VCU118), Zynq-7 (ARM+FPGA), Kintex-7, TI's TDA2SX, AWR1843 (ARM+DSP), Jetson Xavier NX, AGX Orin

Interfaces and Protocols:

SpaceWire (RMAP, CCSDS), PCIe (Endpoint and Root Complex), MIPI CSI-2 (D-PHY), Serial LVDS, AXI4-Stream/Memory-Map, APB Bus, Ethernet, SPI/QSPI/I2C/UART/CAN FD, JESD204B

EDA and Software Tools:

Libero SoC, Synplify, Vivado, Questa/ModelSim, Polyspace, MATLAB, and Eclipse based IDEs

Embedded and Linux:

C/C++, Yocto Linux Project, kernel drivers/device trees, bare-metal, FreeRTOS, TI-RTOS, Git; VxWorks (bring-up)

Standards:

ECSS-E-ST-50-12C / 51C / 52C / 53C (Aerospace), ISO 26262-6 (Automotive Functional Safety), MISRA C, IEEE 1735 (IP Encryption)

Employment History

FPGA/Embedded Systems Engineer

Sep 2025 - Present

CAVU Aerospace UK

Own the architecture and development of FPGA RTL and embedded software for high-speed interfaces across CAVU's camera, onboard-computer (OBC), and edge-computing products in a fast-paced aerospace startup. Involved in meeting with customers for technical discussions and product feasibility. Early evaluation of upcoming PIC64-HPSC.

- Designed and implemented an end-to-end SpaceWire subsystem for OBC product family, supporting hardware-accelerated RMAP and CCSDS-over-SpaceWire protocols. The design included distributed reset subsystems for processor interconnection, SpW CODEC, and DMA units/buffers for fault tolerance and auto recovery across multiple clock domains. Runtime configurations were supported via AXI4-MM access to control units.
- Designed and implemented a multi-descriptor scatter-gather DMA engine. The DMA was specifically designed to accommodate the high, non-deterministic latency of Linux by supporting descriptor exhaustion and multiple in-flight packets. Link errors, timeouts, and packet drops were handled with automatic recovery and error reporting to the host. The DMA engine was integrated with a 3-camera setup (SpaceWire), enabling simultaneous frame transfer of up to 18.6 MB/s per camera (link rate: 200 Mb/s).
- Implemented a frame-transfer pipeline from the camera sensor (post-ISP) through MIPI CSI-2, supporting 1-, 2-, and 4-lane configurations at up to 4 Gb/s with HS-IO and 10 Gb/s with transceivers. The pipeline included a control plane shared between Jetson (via I2C) and RISC-V (via APB). The I2C control plane offloaded camera control from the main processor, enabling low-latency operation without user intervention.
- Conducted feasibility tests for six 1-Gb Ethernet nodes on PolarFire SoC, comparing XCVR/SGMII/PCDMA against a PCIe switch. Early testing of these architectures informed the final design choice.
- Developed board support packages, including device trees and kernel drivers, using the Yocto Linux Project, JetPack for Jetson platforms, and bare-metal drivers.
- Performed software bring-up of the PIC64-HPSC, beginning hardware-capability testing with QEMU simulation.
- Created and presented technical content for Microchip workshops on behalf of CAVU Aerospace.

FPGA/Software Developer

Jun 2024 - Aug 2025

Institute of Communication Systems, University of Surrey

Worked as part of a research-focused team on a UKRI-funded project, implementing and evaluating a novel non-linear MIMO detection algorithm on FPGA hardware accelerators and NXP LA12xx platforms, including integration with OpenAirInterface (OAI).

- Designed high-throughput datapath for matrix operations and LLR estimation (Detection Unit), including a ping-pong memory architecture and sequencing logic to minimize accelerator idle time.
- Refined the detection-unit microarchitecture, reducing FPGA resource utilization. By restructuring DSP slices as 3-input MAC, and replacing a fully parallel MAD tree with a pipelined sort-and-search, reduced fabric and DSP resources by 50%. Allowing both pre and post-processing to fit on a single VCU118, whereas the original design required two.
- Integrated the hardware accelerator with host software (OAI) using PCIe interface (XDMA). Pipelined host-to-card and card-to-host transfers so the next input batch was transmitted while the current batch's results were returned, overlapping PCIe latency.
- Implemented a custom soft-input, soft-output LDPC decoder targeting high throughput for integration with the MIMO detection pipeline.
- Mapped the detection algorithm on two contrasting architectures: a floating-point SIMD/VLIW implementation on the NXP VSPA and RTL datapath on FPGA, evaluating implementation complexity, throughput, and latency.

FPGA/Embedded Engineer and Team Lead

Dec 2020 - Mar 2024

Sarmad Communication Technologies

Joined the company as its first employee and led the development of FPGA and embedded SoC systems for forward-collision radar applications. During my tenure, the company grew from 3 to more than 20 employees, while I led a team of 6 embedded and systems engineers.

- Defined datapath architectures, control logic, and interface and protocol requirements based on system-level specifications. Followed a V-model process, defining hardware-software partitioning and breaking tasks for the team.
- Designed and implemented processing pipelines on Zynq-7000, implementing PL logic in VHDL and PS software in C with high-throughput data handling through external DDR memory.
- Developed embedded C/C++ software for forward-collision radar using TI's TDA2SX ADAS processor and AWR1843 SoC, covering signal processing, data processing, and platform integration following ISO 26262 guidelines.
- Implemented a serial LVDS ADC interface using IDELAY and ISERDES primitives and developed timing constraints for reliable data capture. The ADC was a 16 channel, 14-bit, at sampling rate of 50 MSPS.
- Collaborated with PCB and RF teams on schematic reviews and part number selection.

FPGA Engineer

Dec 2018 - Nov 2020

Middle East Pioneer Communication

Hardware engineer in charge of development of FPGA-based signal processing units and high-speed data acquisition interfaces, including RTL design in VHDL and verification in Verilog.

- Integrated high-speed ADCs using JESD204B at 12.5 Gbps and parallel LVDS interfaces at 250 MSPS. Working with these ADCs strengthened my understanding of timing and clocking issues. I resolved data-integrity problems through appropriate timing constraints and floorplanning, achieving reliable data capture and timing closure.
- Implemented signal processing units using mixed fixed/floating-point arithmetic on Kintex-7. The switch to floating-point arithmetic actually saved resources by 20% without compromising accuracy.
- Design and integration of communication interfaces including Gigabit Ethernet, UART, SPI, QSPI, and memory controllers (SRAM, NAND Flash).

Education/Training

B.Sc. in Electrical Engineering, major in Digital Systems - University of Tehran

2015-2019

*Bachelor's Project: **Implementation of CNN Network with Design Space Exploration in Low-Cost FPGAs***

Advanced Vivado FPGA workshop - Yasin Engineering Co

2020-2021

*Covering subjects such as TCL scripting, floorplanning, timing analysis, and power optimization in Vivado.
Completed with grade A*